A Point-of-Care, Real-Time Artificial Intelligence System to Support Clinician Diagnosis of a Wide Range of Skin Diseases

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Dermatological diagnosis remains challenging for nonspecialists because the morphologies of primary skin lesions widely vary from patient to patient. Although previous studies have used artificial intelligence (AI) to classify lesions as benign or malignant, there have not been extensive studies examining the use of AI on identifying and categorizing a primary skin lesion's morphology. In this study, we evaluate the performance of a standalone AI tool to correctly categorize a skin lesion's morphology from a test bank of images. To provide a marker of performance, we evaluate the accuracy of primary care physicians to categorize skin lesion morphology in the same test bank of images without any aids and then with the aid of a simple visual guide. The AI system achieved an accuracy of 68% in determining the single most likely morphology from the test image bank. When the AI's top prediction was broadened to its top three most likely predictions, accuracy improved to 80%. In comparison, the diagnostic accuracy of primary care physicians was 36% without any aids and 68% with the visual guide ($P < 0.001$). The AI was subsequently tested on an additional set of 222 heterogeneous images of varying Fitzpatrick skin types and achieved an overall accuracy of 70% in the Fitzpatrick I–III skin type group and 68% in the Fitzpatrick IV–VI skin type group ($P = 0.79$). An AI is a powerful tool to assist physicians in the diagnosis of skin lesions while still requiring the user to critically consider other possible diagnoses.

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INTRODUCTION

There are more than 3,000 known skin diseases that include neoplasms, autoimmune reactions, and inflammatory eruptions that are either primarily dermatological or manifestations of internal disease (Lim et al., 2017). The presentation of skin disease varies widely across skin types, disease acuity, immune status, and treatment history. Thus, dermatological diagnosis remains challenging, particularly for nonspecialists. Most medical schools offer less than 10 hours of total dermatology training (McCleskey et al., 2009). Nondermatologists correctly diagnose skin disease in <50% of cases. Given that nearly a quarter of all visits to a healthcare provider involve a skin-related problem (Lim et al., 2017), there is a need for scalable clinical decision tools to support diagnosis, triage, and treatment decision-making for nondermatologists. The broad range of possible skin diseases and presentations of those diseases makes this challenge particularly well suited for applications of artificial intelligence (AI)-based clinical decision support tools.

Given the visibility of skin disease, dermatology has remained an active area for AI research in medicine. Although there have been Food and Drug Administration–approved systems for the assessment of skin lesions (MelaFind), the technology never gained traction within the clinical community and has been discontinued by the manufacturer (Monheit et al., 2011). More recent research has leveraged AI—focused on classifying skin neoplasms—with greater specificity using histopathology or dermatologist diagnosis as ground truth (Esteva et al., 2017; Marka et al., 2019; Rajpara et al., 2009). Although AI has utility for clinicians evaluating neoplasms, skin biopsy remains a low-risk procedure on a highly accessible organ that offers a histopathological level of diagnostic certainty. Thus, the major unmet clinical need for AI in dermatology lies in the need for explainable, nonbiased, point-of-care systems able to guide nonexpert clinicians through a wide dermatological differential diagnosis—including solitary lesions and rashes where histology is considerably less definitive. Fundamental to the generation of a dermatological differential is first to classify correctly the primary and secondary morphological features (e.g., patch, plaque, papule, ulcer with or without scale) of any skin disease—a task that can be facilitated by clinical decision support via AI tools (Burgin, 2019).

In this study, we report the development and early validation of a point-of-care, real-time, and generalizable AI system trained on 76,926 dermatologist-labeled images specifically designed to guide a clinician through a dermatological differential diagnosis. We provide context on the performance of this AI system by comparing the classification accuracy of 15 board-certified internal medicine physicians with and without a visual aid of skin disease morphologies.

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Abbreviation: AI, artificial intelligence

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We further report this system's performance on an additional test image set of 222 images of a wide range of skin disease morphologies and skin Fitzpatrick types.

RESULTS

Primary care physicians (n = 15) were asked to provide morphology descriptions of 16 clinical images (Figure 1). Clinical experience of the physicians ranged from 1–4 years (n = 6), 5–9 years (n = 3), 10–14 years (n = 1), and 15–19 years (n = 5). These 15 participants achieved a median baseline score of 36% (9 of possible 25 points) on the test set of 16 images (Figure 2). Participants then repeated the same exercise with the assistance of a visual guide of dermatological morphologies and achieved a significantly improved median score of 68% (17 of possible 25 points, $P < 0.001$). The AI system was then deployed on the same 16 test images and scored 68% with its most confident classification, equivalent to the median score of study participants using a visual aid. When broadening to include the AI's top three classifications, the AI scored 80%, scoring better than 14 out of 15 study participants.

The AI algorithm was applied subsequently to a larger set of 222 images, including a wider range of skin pathologies and skin types (Figure 3). Morphologies in this larger test set and the associated percentage correctly predicted by the AI tool's most confident prediction (across all Fitzpatrick skin types) include: plaque (n = 64, 72%), papule (n = 42, 64%), patch (n = 22, 73%), ulcer or erosion (n = 12, 75%), cyst (n = 9, 89%), scar (n = 8, 38%), bullae (n = 7, 57%), purpura (n = 6, 83%), pustule (n = 6, 83%), vesicle (n = 5, 60%), macule (n = 3, 33%), nodule (n = 3, 100%), and other lesions (n = 35, 69%). The classifier's overall accuracy across all 222 images was 69%; when considering the AI tool's top three predicted classifications—assessing for a correct response among its top three predictions—accuracy improved to 92% across all images. Given the range of hundreds of possible morphological assignments the AI could select from, an accuracy of less than 1% would be expected if chosen by chance alone.

The Fitzpatrick skin type for all 222 images was recorded as one of two groups: Fitzpatrick types I–III (n = 169) or Fitzpatrick types IV–VI (n = 53). The AI accuracy for each morphology was stratified by Fitzpatrick skin type (Figure 3). The overall accuracy in the Fitzpatrick I–III skin type group was 70% for the most confident prediction, whereas accuracy in the Fitzpatrick IV–VI skin type group was 68%; the difference in performance for the AI most confident prediction was not statistically significant ($P = 0.79$, Figure 4). When assessing the AI tool's top three predictions, accuracy in the Fitzpatrick I–III skin type and Fitzpatrick IV–VI skin type groups improved to 92% and 92%, respectively; no statistically significant difference in performance exists between the Fitzpatrick skin type groups for the top three predictions ($P = 0.86$).

DISCUSSION

The results from the initial primary lesion test indicated that for this cohort of physicians, the baseline accuracy at identifying skin lesion morphologies was 36%, which is concordant with the performance levels reported previously (Ahiarah et al., 2007; Bukhari and AlAkloby, 2006; Sherertz, 1990). This demonstrates a clear need for diagnostic tools to improve clinician skills in accurately describing and diagnosing skin lesions. Accuracy of lesion description within our test cohort improved to 68% ($P < 0.001$) when primary care physicians were given a simple visual guide with no additional training. The data suggest that a static visual guide can improve morphological classification of dermatological conditions by physicians. This tool will allow physicians to provide more accurate descriptions of their patients' skin lesions to consulting dermatologists when needed, which may help to facilitate more rapid care decisions within a busy clinic schedule. However, a static visual guide requires additional time and attention from physicians, which may interrupt the workflow of a typical physician-patient encounter (Alther and Reddy, 2015). A mobile device–deployable AI tool that automatically classifies skin morphologies from neoplasms to rashes offers a streamlined, point-of-care option that is now commercially available (VisualDx DermExpert, Rochester, NY).

Overall, the AI tool offered a high level of accuracy when tasked with identifying a broad set of dermatological images ranging from inflammatory to neoplastic conditions and including a broad morphological corpus. Although several previous studies have focused on determining the likelihood of malignancy from a photographed lesion (Esteva et al., 2017), none have attempted to categorize the morphology of the primary lesion across the broad presentation range of skin disease with a point-of-care tool. In addition, the images were taken from real-world clinical encounters and included large variances in lighting, camera models, image angles, and possible distractors such as patient clothing or jewelry, which emphasizes the ease of use of the AI tool.

There has been recent concern that AI will further exacerbate healthcare disparities and inequality between patients of different skin color (Adamson and Smith, 2018; Char et al., 2018; Rajkomar et al., 2018). It is crucial to recognize that as machine learning becomes more advanced and prevalent in clinical medicine, physicians and researchers must be cognizant of how a particular AI is built. The validation of AI in a wide variety of patients is vital to ensure both adequate real-world clinical performance and that vulnerable populations are not excluded with its use. When this AI was tested on a larger set of 222 images, there was no statically significant difference in its rates of accuracy when given a Fitzpatrick I–III versus Fitzpatrick IV–VI skin type image, suggesting that the model is equally accurate among heterogeneous skin tones and colors.

Another significant challenge of AI across all fields is that, over time, users of AI will become overreliant on an AI system and fail to gain personal expertise in critical thinking tasks (e.g., deskilling). As an example, pilots who constantly use autopilot functionality might lose the skills to fly the plane manually in a wide range of circumstances (Arnold, 2015). In medicine, this could mean doctors overrelying on AI to tell them the diagnosis without thinking through and confirming the accuracy of the diagnosis based on pathophysiologic understanding and medical logical competence. A related challenge is that machine learning of an image will never be 100% accurate and the professional

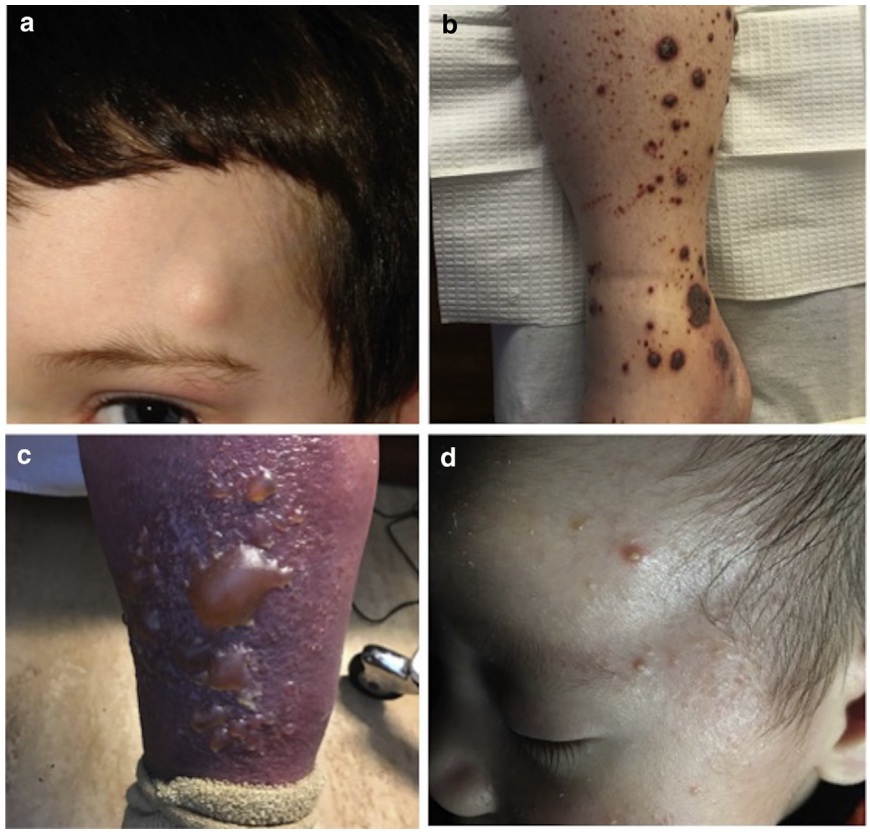


Figure 1. Sample clinical images from the 16-image set used to test both internal medicine physicians and the AI system. Physician participants were asked to describe each image in three words or fewer. Points were awarded if the answer included the correct term or terms. Correct answers for the image set are (**a**) nodule (1 point), (**b**) purpura (1 point) and palpable or papule (1 point for either), (**c**) bullae (1 point), and (**d**) pustule (1 point). Written consent was provided for use of de-identified images by patients, or in the case of minors, a parent. AI, artificial intelligence.

will need to understand the inherent limitations of the system. Adding to this concern is the fact that dermatological differential diagnosis is not always made through pattern recognition; medical history is often critical in making an accurate diagnosis (Papier, 2012). For example, it is nearly impossible to attribute the etiology of a bullae to bullous pemphigoid or a friction blister relying on an image alone. The patient demographics and history are important to distinguish these conditions even with an identical visual presentation. Although laboratory assessment and skin biopsy for histopathological study can aid in the diagnosis, these tests cannot always be easily performed in the primary care setting and can take hours to days to complete in the emergency or inpatient setting (Santistevan et al., 2017). Especially in the case of life-threatening conditions, a morphology-driven differential diagnosis can guide early treatment decisions and determine the next best step in work-up (Jones and Walling, 2007). A recent study has demonstrated that morphology-based clinical decision support software can augment a general practitioner's diagnostic accuracy (34%; $P = 0.007$) in real-world clinical scenarios with greater patient satisfaction and without increasing the length of consultation (Breitbart et al., 2020). For this reason, an AI tool that provides morphological assessment may help users build a differential diagnosis tailored to that individual patient and would be of clinical use in the primary care and emergency settings. A well-trained provider can develop a focused differential diagnosis with a combination of accurate clinical interpretation and a holistic understanding of their patient. Future efforts should be made to further validate whether these systems improve diagnostic accuracy in more real-world patient interactions.

A limitation of this study is that the diagnostic tool and AI were focused on primary lesions rather than direct diagnoses.

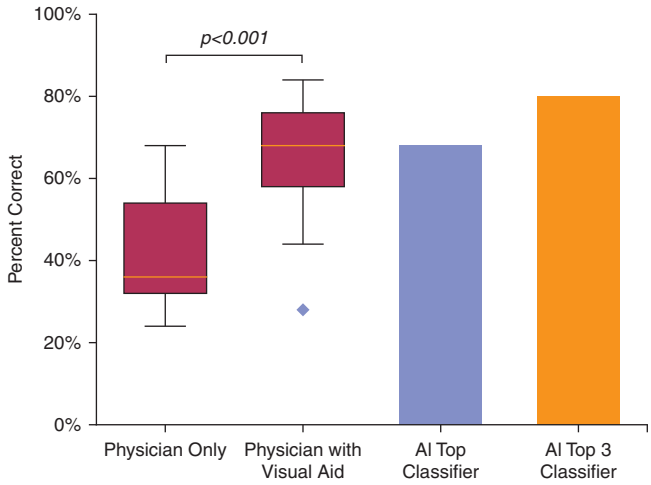


Figure 2. Performance on the 16-image test set. Primary care physicians performance improved from an average of 36% to 68% with the addition of a simple visual guide ($P < 0.001$ by Kruskal-Wallis test for rank sum difference). The AI tool provided both a top classifier, or best guess, and top three classifiers, or best three guesses. The top classifier matched that of primary care physicians using a visual guide (68% correct), and the top three classifiers included the correct answer 80% of the time. AI, artificial intelligence.

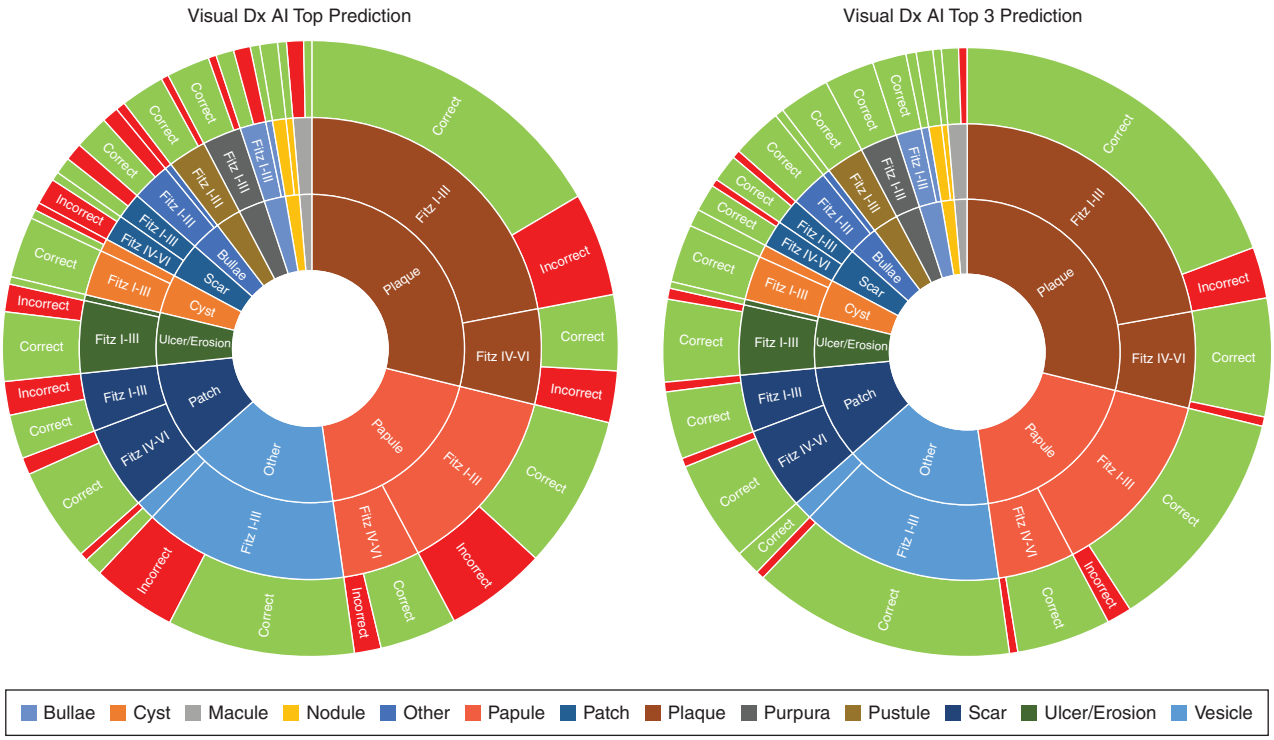


Figure 3. Performance of the AI algorithm on the 222-image test set. Sunburst plots of the AI algorithms top classifier (left) and top three classifiers (right) demonstrate the breakdown of the 222-image set by morphology (inner ring), Fitzpatrick type (middle ring), and percent correct (outer ring). AI, artificial intelligence.

Thus, the utility of this system is highest in instances where a diagnosis is not immediately clear to the provider. By design, the system development strategy is to combine multiple machine learning models with clinical decision support rules. The complexity of broad dermatological diagnosis across hundreds or thousands of diagnoses means there is a critical need to engineer thinking checks in the AI process. An additional limitation in this study is that the AI was only tested on 222 images of different Fitzpatrick skin types. Further work of combining diagnostic machine learning models, skin lesion morphology models, and a clinical decision support engine should expand performance and reliability. Finally, in creating our test set for our physician participants, images were chosen by a group of three dermatologists to represents a wide range of skin disease morphologies. As this was done nonrandomly, this is a potential source of selection bias.

The American Academy of Dermatology's position statement on augmented intelligence provides guidelines on the future of this technology in clinical practice that highlight the

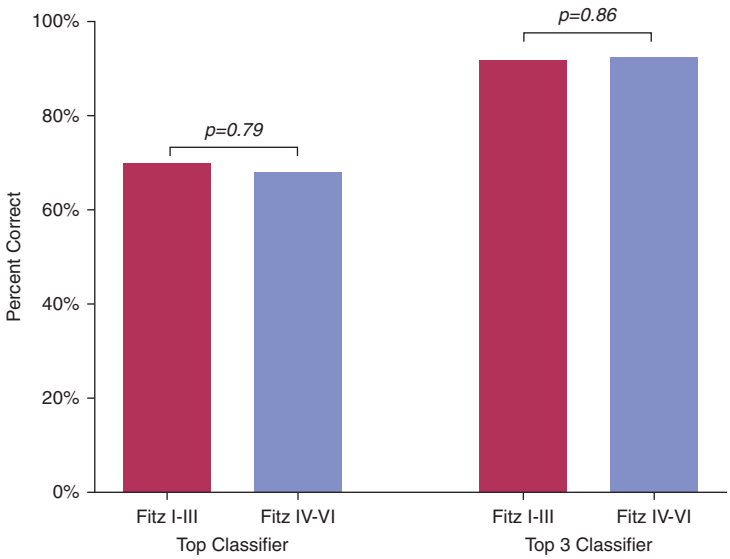


Figure 4. No difference in performance of the AI algorithm by Fitzpatrick type. The overall accuracy of the top classifier was 70% in the Fitzpatrick I–III skin type group and 68% in the Fitzpatrick IV–VI skin type group ($P = 0.79$ by Student's *t*-test). The overall accuracy of the top three classifiers increased to 92% in the Fitzpatrick I–III skin type and 92% in the Fitzpatrick IV–VI skin type groups ($P = 0.86$). AI, artificial intelligence; Fitz, Fitzpatrick.

importance of the quality of training data, careful clinical deployment with appropriate real-world surveillance, and the need for an educated clinician workforce on the advantages and limitations of AI (American Academy of Dermatology, 2019). Although direct diagnosis is an eventual goal of AI, the step of making sure the user confirms a correctly identified primary lesion as with this AI algorithm provides two distinct benefits: (i) it acts as a quality control check and (ii) it simultaneously educates the non-dermatologist in the language of skin exam morphology. These benefits produce transparency for patients and ensure that the ultimate clinical decision-making rests with a physician and not a model.

MATERIALS AND METHODS

Skin lesion description by internal medicine physicians

A panel of 16 de-identified clinical images taken from several smartphones was selected by three board-certified dermatologists (BD, MC, and SX) to display a range of skin disease morphologies that might commonly be seen in a primary care setting. All patients or their parent or guardian provided written, informed consent for the use and publication of de-identified clinical images. A key was generated and reviewed by all authors using a 1−2 word description of each image, including a primary morphology with or without a secondary morphology (Supplementary Files 1 and 2). Primary care physicians (board certified in internal medicine) were recruited (n = 15) and asked to describe the dermatological lesion shown in each image in free text that would effectively communicate their perceived morphology. This study was deemed exempt from approval by the Northwestern University Institutional Review Board. To score their responses, one point was given for each descriptor matching the key, with a maximum of 25 points possible. After completing the survey without assistance, participants were asked to complete the same test images with a visual aid and legend describing and defining various morphologies. Correct responses for each test image were determined by three dermatologists (BD, MC, and SX), with an interrater reliability of 100%.

Skin lesion description by AI tool

The AI tool used in this study, available for use (VisualDx DermExpert), was trained on over 76,926 dermatologist-labeled images from the VisualDx privately curated image database (69,195 training images, 3,869 validation images, and 3,862 test images). The core algorithm relies on an ensemble of five trained convolutional neural network models. These five modules are based on well-known and validated convolutional neural network models previously reported in the AI literature, including DenseNet (Huang et al., 2019) and NASNetMobile (Zoph et al., 2018), as well as proprietary models developed by VisualDx. Each module has a distinct architecture and was trained to different parameters. To process a new dermatological image using the ensemble, the image is run through each model separately, and results are then averaged across all models. This process results in a single best classifier, which is the most likely morphology as determined by the AI, and two other possible morphologies that it has deemed possible but less likely (forming its top three morphological assessment). The single best classifier revealed that an ensemble technique yielded an absolute 7.2% and 12.3% relevant improvement compared with the single best model (58.3% accuracy for ensemble vs. 51.1% for single best model). For the top three results, the ensemble method yielded the best performance (82.2%) compared with the single best model (76.7%), which represents a 5.5% absolute improvement and a 6.6% relative improvement.

The AI tool was then used to independently categorize the same panel of 16 images tested on physician participants. None of these images were used previously to train the model. The tool outputs included a single, highest-confidence classification and its top three morphological classifications, in order of confidence.

To further assess the performance of the AI tool, we evaluated the performance of the system in characterizing morphologies in an additional 222-image set representing a wide range of real-world clinical images from the authors' personal collections and contributions from outside dermatologists from a diverse patient population. This set of images was novel to the AI tool and was not used previously for AI training. These images were selected and reviewed by two dermatologists (BD and SX). Ground truth morphologies were assigned to these images by two dermatologists (BD and SX), who achieved an interrater reliability of 100%. Images were taken from a range of smartphone cameras and no modifications were made to the images or backgrounds. Images included in this dataset represent a wide range of skin pathologies, including 29% plaques; 19% papules; 10% patches; 5% erosions or ulcers; and less than 5% of bullae, cysts, macules, nodules, purpura, pustules, scars, vesicles, and other lesions. In addition, images were classified by Fitzpatrick skin type and separated into darker skin types (Fitzpatrick skin type IV−VI) and lighter skin types (Fitzpatrick skin type I−III). The AI's output was graded by BD and SX and any discrepancies were discussed until consensus was reached; grades were stratified by lesion type and Fitzpatrick skin type, and performance of both highest-confidence classification and top three classifications were considered.

Statistical methods

Performance of internal medicine physicians in assessing morphology of the 16-image test set was compared before and after use of a visual aid with a Kruskal-Wallis test for ranked sum difference. AI performance was compared between lighter and darker skin types (Fitzpatrick types I−III and types IV−VI, respectively) across the highest-confidence classification and top three classifications with a Student's *t*-test. All statistical analysis was conducted in Rstudio version 1.1.453 using R version 3.5.1.

Data availability statement

No datasets were generated or analyzed during the current study.

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CONFLICT OF INTEREST

SX reports a prior paid consultant role with VisualDx that ended December 31, 2017. He has no active conflict of interests that relate to this work. The remaining authors state no conflict of interest.

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AUTHOR CONTRIBUTIONS

Conceptualization: BD, SX; Data Curation: BD, MC, SX; Formal Analysis: KT, BD, SX; Writing - Original Draft Preparation: KT, MZZ, BD, SX; Writing - Review and Editing: BD, SX, MC, KT, MZZ

SUPPLEMENTARY MATERIAL

Supplementary material is linked to the online version of the paper at www.jidonline.org, and at https://doi.org/10.1016/j.jid.2020.08.027.

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